Karim ElSedfy

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AI/ML Research · Generative Modeling · Computer Vision

Education

Bowdoin College May 2028
B.A. in Computer Science and Mathematics *Brunswick, ME*
- **GPA:** 3.85/4.00, Dean's List
- **Relevant Coursework:** Deep Learning, Artificial Intelligence, Computational Complexity, Algorithms, Computer Systems, Linear Algebra

Research & ML Experience

Computer Science Department, Bowdoin College Sep. 2025 – Dec. 2025
Independent Study Researcher *Brunswick, ME*
- Investigated whether a frozen, detection-trained YOLO encoder could transfer to semantic segmentation, implementing dense stride-1 inference and controlled decoder variants with and without U-Net skip connections in PyTorch.
- Improved mean IoU from 0.600 to 0.684 (+14.0%) on CUB-200 while preserving 0.88 detection mAP@0.5; analyzed the ablation results and authored an IEEE-format paper under Prof. Jeova Farias.

Palmera Systems Co. Jun. 2025 – Aug. 2025
Software Engineer Intern *Cairo, Egypt*
- Architected face recognition and anti-spoofing for the smartIN app (8K daily users), leading dataset curation, model fine-tuning, and evaluation with DeepFace/OpenCV to reach 96% detection accuracy.
- Deployed the models behind PostgreSQL-backed REST services matching live captures in under 2s, cutting authentication latency by 50%.

Ungated Research, Bowdoin College Mar. 2026 – Jun. 2026
Software Engineer *Brunswick, ME*
- Co-maintained a multithreaded Python research-paper pipeline across 11+ economics journals, adding LLM-generated summaries, tags, and translations plus error-detection and automated-repair diagnostics; surfaced 6,080 new articles through 10 years of Crossref records.

Dropbox May 2026 – Aug. 2026
Software Engineer Intern *Remote – San Francisco, CA*
- Won two of three company hackathon awards leading an AI system that converted customer reports into privacy-safe engineering investigations by correlating product activity, production errors, code changes, and issue history.
- Shipped two full-stack experiments reaching ∼350K users weekly; the first launched four weeks early and the second lifted Standard-plan conversion 2.3× over the prior flow.

Selected AI Projects

Persona-Motion: Audio-Driven Talking-Head Synthesis GitHub
Python, PyTorch, Flow Matching, LivePortrait, Slurm HPC
- Reframed audio-to-motion prediction as rectified flow matching after diagnosing mean collapse in an MSE baseline; a 14.7M-parameter BiGRU velocity field maps wav2vec2 features to 205-d facial-motion coefficients rendered by frozen LivePortrait.
- Designed the training and evaluation pipeline on a Slurm GPU cluster, including classifier-free guidance and a from-scratch contrastive SyncNet evaluator; improved lip-sync confidence ∼6× (0.09 → 0.56, vs. 0.77 ground truth).

Lemma Notes AI: Handwriting-Native AI Notebook GitHub
Swift, Vision, PencilKit, Gemini, Cloud Run, Firestore
- Built an on-device handwriting synthesis system from first principles: guided glyph calibration, continual sample harvesting, chamfer-distance stroke classification, baseline-aware layout, variation sampling, and pen-pressure matching.
- Designed a hybrid Vision OCR + Gemini pipeline where hosted models return only schema-constrained JSON and deterministic code validates, places, and renders ink; verified the 40K-line app/server system with 960+ automated tests.

Technical Skills

Languages: Python, Swift, TypeScript, JavaScript, SQL, Java
ML / Research: PyTorch, flow matching, classifier-free guidance, Hugging Face Transformers, OpenCV, Vision OCR, model evaluation, ablation studies
Systems / Data: Slurm HPC, Docker, Kubernetes, Cloud Run, AWS S3, Firestore, PostgreSQL, Redis, SQLAlchemy
Frameworks and Tools: SwiftUI, PencilKit, React, FastAPI, Node.js, Git, GitHub Actions